Practice Problems

1. Decision Tree: Overfitting vs Underfitting

Dataset: Iris Dataset (UCI/Kaggle)

Download:

- https://www.kaggle.com/datasets/uciml/iris

Tasks

1. Load the dataset.
2. Split it into 80% training and 20% testing data.
3. Train two Decision Tree models:
   - Model A: `max_depth=2`
   - Model B: `max_depth=None`
4. Compare:
   - Training Accuracy
   - Testing Accuracy
5. Which model is underfitting? Which one is overfitting?

2. Finding the Best Tree Depth

Dataset: Wine Dataset

Download:

- https://www.kaggle.com/datasets/uciml/red-wine-quality-cortez-et-al-2009

Tasks

1. Load the dataset.
2. Train Decision Trees using:
   - depth = 2
   - depth = 4
   - depth = 6
   - depth = 8
   - depth = None
3. Create a table containing:
   - max_depth
   - Train Accuracy
   - Test Accuracy
4. Which depth gives the best generalization?

3. Random Forest vs Decision Tree

Dataset: Breast Cancer Wisconsin Dataset

Download:

- https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

Tasks

1. Train a Decision Tree.
2. Train a Random Forest (`n_estimators=100`).
3. Compare:
   - Accuracy
   - Precision
   - Recall
4. Which model performs better and why?

4. Effect of Number of Trees

Dataset: Heart Disease Dataset

Download:

- https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

Tasks

Train Random Forest with:

- 10 trees
- 50 trees
- 100 trees
- 200 trees

Create a comparison table of:

- Number of Trees
- Train Accuracy
- Test Accuracy
- Training Time

Which number of trees gives the best balance between performance and computation?

5. Understanding Bias-Variance Tradeoff

Dataset: Titanic Dataset

Download:

- https://www.kaggle.com/competitions/titanic/data

Tasks

Train the following models:

- Decision Tree (depth=2)
- Decision Tree (depth=10)
- Random Forest (100 trees)

Compare:

- Training Accuracy
- Testing Accuracy

For each model determine:

- High Bias?
- High Variance?
- Balanced?

Explain your reasoning.

6. Visualizing Overfitting

Dataset: Pima Indians Diabetes Dataset

Download:

- https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Tasks

Train Decision Trees for every depth from:

1 to 20

Create a graph:

- X-axis → Tree Depth
- Y-axis → Accuracy

Plot both:

- Training Accuracy
- Testing Accuracy

Questions:

- At which depth does overfitting begin?
- Why?

---

7. Hyperparameter Tuning of Random Forest

Dataset: Adult Income Dataset

Download:

- https://www.kaggle.com/datasets/uciml/adult-census-income

Tasks

Perform Grid Search on:

n_estimators:
50
100
200

max_depth:
5
10
20
None

max_features:
sqrt
log2

Find the best model.

Report:

- Best Parameters
- Best Cross Validation Score
- Test Accuracy

8. Feature Importance Analysis

Dataset: Bank Marketing Dataset

Download:

- https://www.kaggle.com/datasets/henriqueyamahata/bank-marketing

Tasks

Train a Random Forest.

Find:

- Top 10 Important Features

Create a bar chart showing feature importance.

Questions:

- Which feature contributes the most?
- Which features contribute the least?
- Remove the least important 5 features and retrain the model.
- Does accuracy change?

9. Decision Tree Pruning for Better Generalization

Dataset: Telco Customer Churn Dataset

Download:

- https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Tasks

Train three Decision Trees:

- No pruning
- `max_depth=6`
- `min_samples_leaf=10`

Compare:

- Training Accuracy
- Testing Accuracy
- Precision
- Recall
- F1 Score

Explain which pruning strategy improves generalization the most.

10. Bias-Variance Analysis Across Multiple Models

Dataset: Dry Bean Dataset

Download:

- https://www.kaggle.com/datasets/muratkokludataset/dry-bean-dataset

Tasks

Train the following models:

- Decision Tree (`max_depth=3`)
- Decision Tree (`max_depth=None`)
- Random Forest (`n_estimators=100`)
- Random Forest (`n_estimators=300`)

Create a comparison table containing:

- Training Accuracy
- Testing Accuracy
- Estimated Bias (High/Low)
- Estimated Variance (High/Low)

Finally, answer:

1. Which model has the highest bias?

2. Which model has the highest variance?
3. Which model generalizes the best?
4. Why does Random Forest usually outperform a single Decision Tree?
5. Explain the relationship between ensemble learning and the Bias-Variance Tradeoff based on your experimental results.